

Lab 1: Set Up the Things

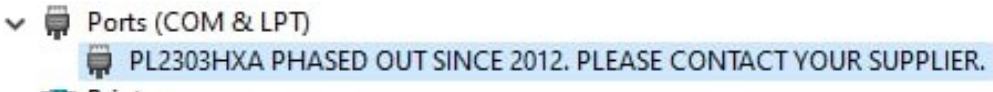
Handout: 29 Sept., Hand-in: 6 Oct 23:59, Edimension

Overview

In this lab, your group will work together to begin the set-ups with your ESP32-CAM and blink the LEDs by your own code:

- You can use any valid GPIO you want.
- You can use either the one-board white light or export GPIO to the LED on the breadboard (GPIO4). CAUTION: please Do not look directly at the white light on board to protect your eyes!
- You may choose either Arduino, MicroPython, or original C++ (<https://docs.espressif.com/projects/esp-idf/en/release-v4.3/esp32/get-started/index.html>) to set up the develop environment.
- If you are using original C++, example source code at esp/esp-idf/examples/get-started/blink/. There may be minor changes in code or configurations, please specify them in your report.

Hints:

1. When setting up the ESP32-CAM, if after connecting the board to your PC with the USB it fails to detect or read the COM port (or get error),
 - please check your device manager under Ports (COM & LPT) if the below appears:

 - or install the older version of the driver (<http://wp.brodzinski.net/2014/10/01/fake-pl2303-how-to-install/>), find a link entitled IO-Cable PL-2303 Drivers-Generic Windows PL2303 Prolific
 - change which driver the USB is using (Right click -> Update driver -> Browse Computer -> Let me pick from drivers on my computer -> pick version 3.3.2.105)
2. When uploading code, remember to hold the reset button for a while.
3. The hardware may not be very stable (e.g., the camera), you can switch to another one, and ask for a replacement from our lab officer, if you found it faulty.

What to hand-in?

1. Demonstrate by presenting photos or a video.

You can either upload it together with your report, or create a link if it is too large. The following should be included in your video:

- To prove that you have successfully installed the environment.
- To present your ESP32 works as expected.

Regarding the hardware: You can use either the ESP32-CAM board or the MakePython development board.

2. Write a brief lab report documenting the following:

- Did you encounter any issue(s) during the setup?
For each issue you encountered, briefly describe the issue and how you solve it.
- Indicate your group ID and all the members' names and student IDs (including audit students, if any) in your lab report.

3. Understand the functions of the components/devices you have (briefly labeled on an image).

Submit your lab report to eDimension before the hand in date. This is a group- based lab and each group just need to submit one lab report.

Total marks for this lab: 25pt

- 10pt: Environment set-ups
- 10pt: Blink the LED
- 5pt: Device Overview